

COMP90042 NLP

L8: Recurrent Networks



Jey Han Lau

Outline

- Recurrent Networks
- Long Short-term Memory Networks
- Applications

N-gram Language Models

- Can be implemented using counts (with smoothing)
- Can be implemented using feed-forward neural networks
- Generates sentences like (trigram model):
 - *I saw a table is round and about*

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 - / *saw a table*

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N-gram Language Models

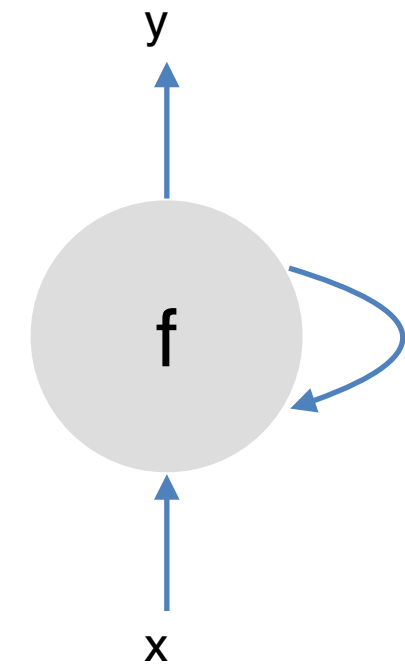
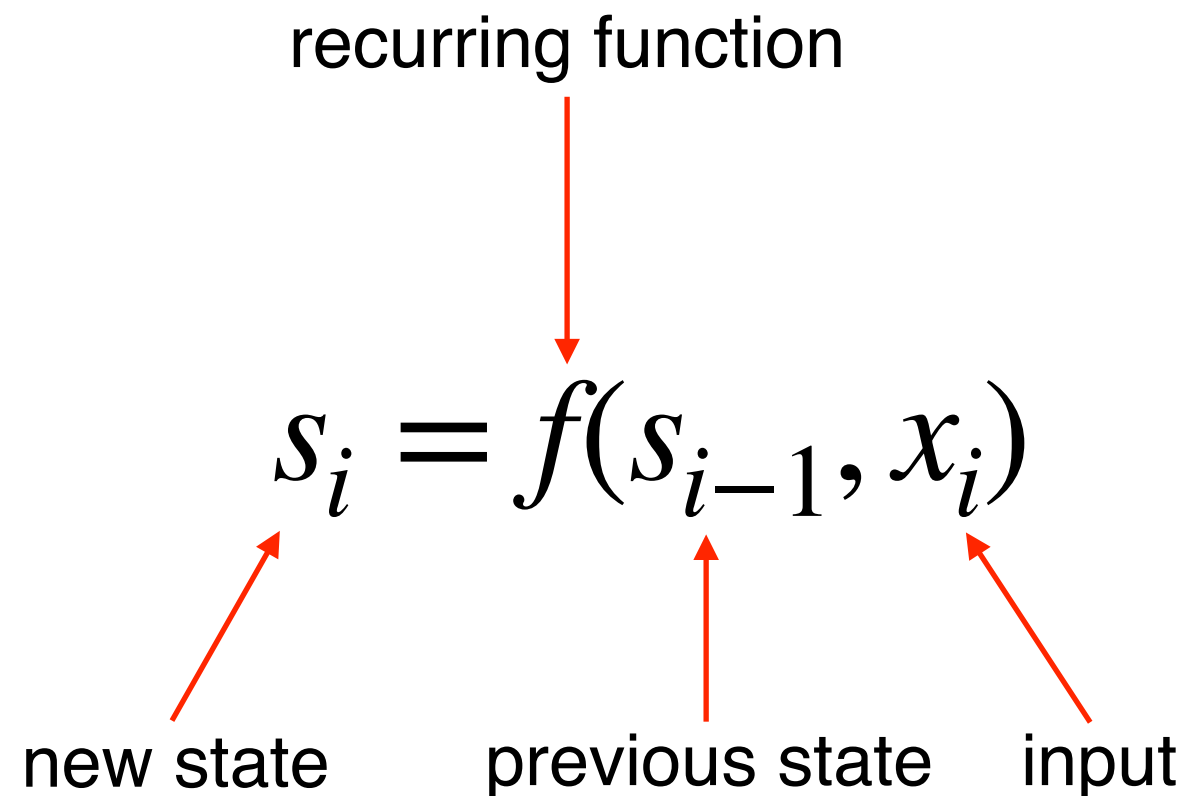
- Can be implemented using counts (with smoothing)
- Can be implemented using feed-forward neural networks
- Generates sentences like (trigram model):
 - *I saw a table is round and about*
- Problem: **limited context**

Recurrent Neural Networks

Recurrent Neural Network (RNN)

- Allow representation of arbitrarily sized inputs
- Core Idea: processes the input sequence one at a time, by applying a recurrence formula
- Uses a **state vector** to represent contexts that have been previously processed

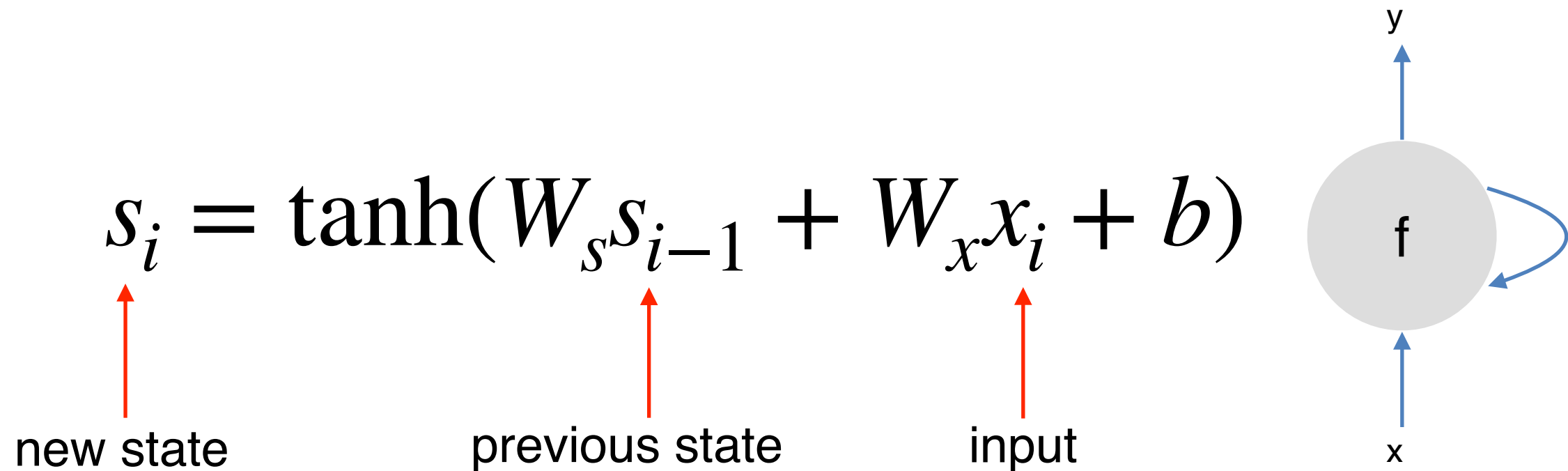
Recurrent Neural Network (RNN)



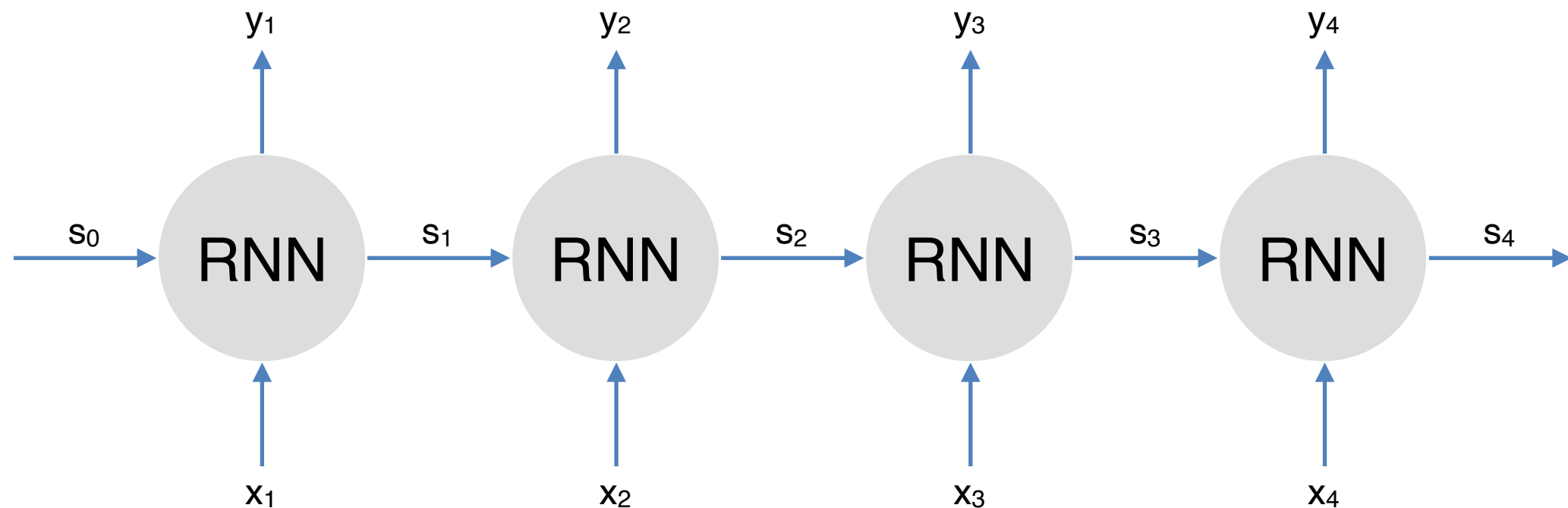
$$s_{i+1} = f(s_i, x_{i+1})$$

same recurring function

Recurrent Neural Network (RNN)



RNN Unrolled



$$s_i = \tanh(W_s s_{i-1} + W_x x_i + b)$$

$$y_i = \sigma(W_y s_i)$$

← “Simple RNN”

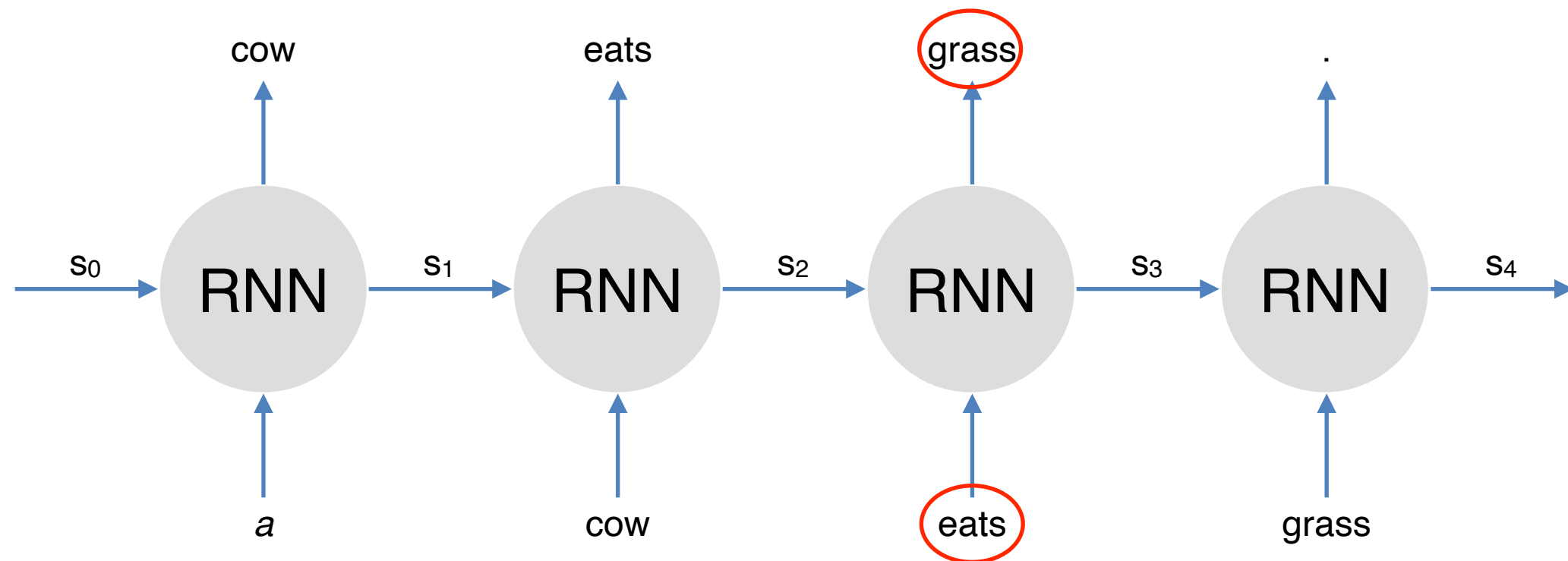
- Same parameters (W_s , W_x , b and W_y) are used across all time steps

RNN Training

- An unrolled RNN is just a very deep neural network
- But parameters are shared across all time steps
- To train RNN, we just need to create the unrolled computation graph given an input sequence
- And use backpropagation algorithm to compute gradients as usual
- This procedure is called **backpropagation through time**

$$\begin{aligned}s_4 &= R(s_3, x_4) \\ &= R(\overbrace{R(s_2, x_3)}^{s_3}, x_4) \\ &= R(\overbrace{R(R(s_1, x_2), x_3)}^{s_2}, x_4) \\ &= R(\overbrace{R(R(R(s_0, x_1), x_2), x_3)}^{s_1}, x_4)\end{aligned}$$

(Simple) RNN for Language Model

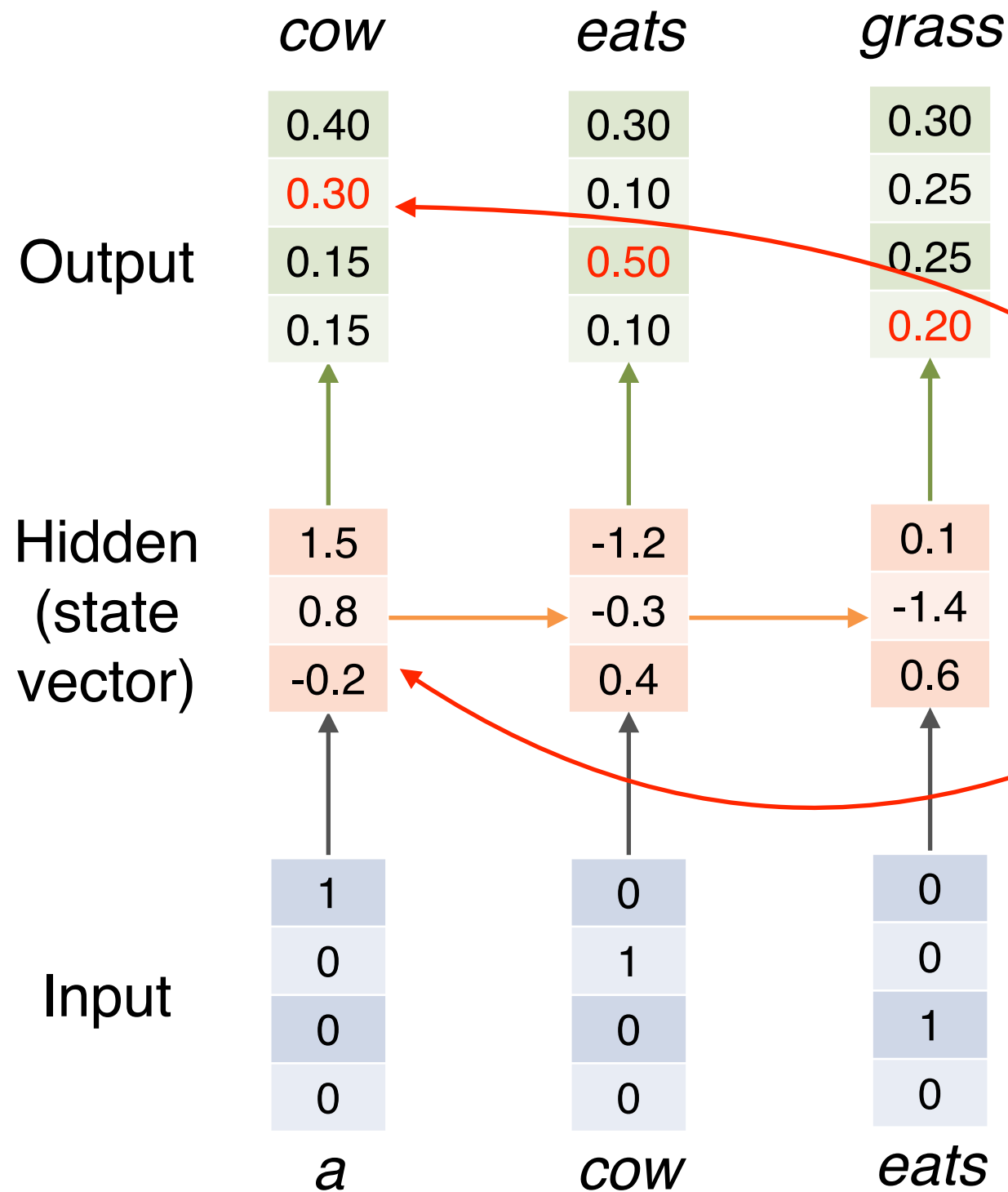


$$s_i = \tanh(W_s s_{i-1} + W_x x_i + b)$$

$$y_i = \text{softmax}(W_y s_i)$$

- x_i is current word (e.g. “eats”); mapped to an embedding
- s_{i-1} contains information of the previous words (“a”, “cow”)
- y_i is the next word (e.g. “grass”)

RNN Language Model: Training



- Vocabulary: [*a*, *cow*, *eats*, *grass*]
- Training Example: *a cow eats grass*

$$s_i = \tanh(W_s s_{i-1} + W_x x_i + b)$$

$$y_i = \text{softmax}(W_y s_i)$$

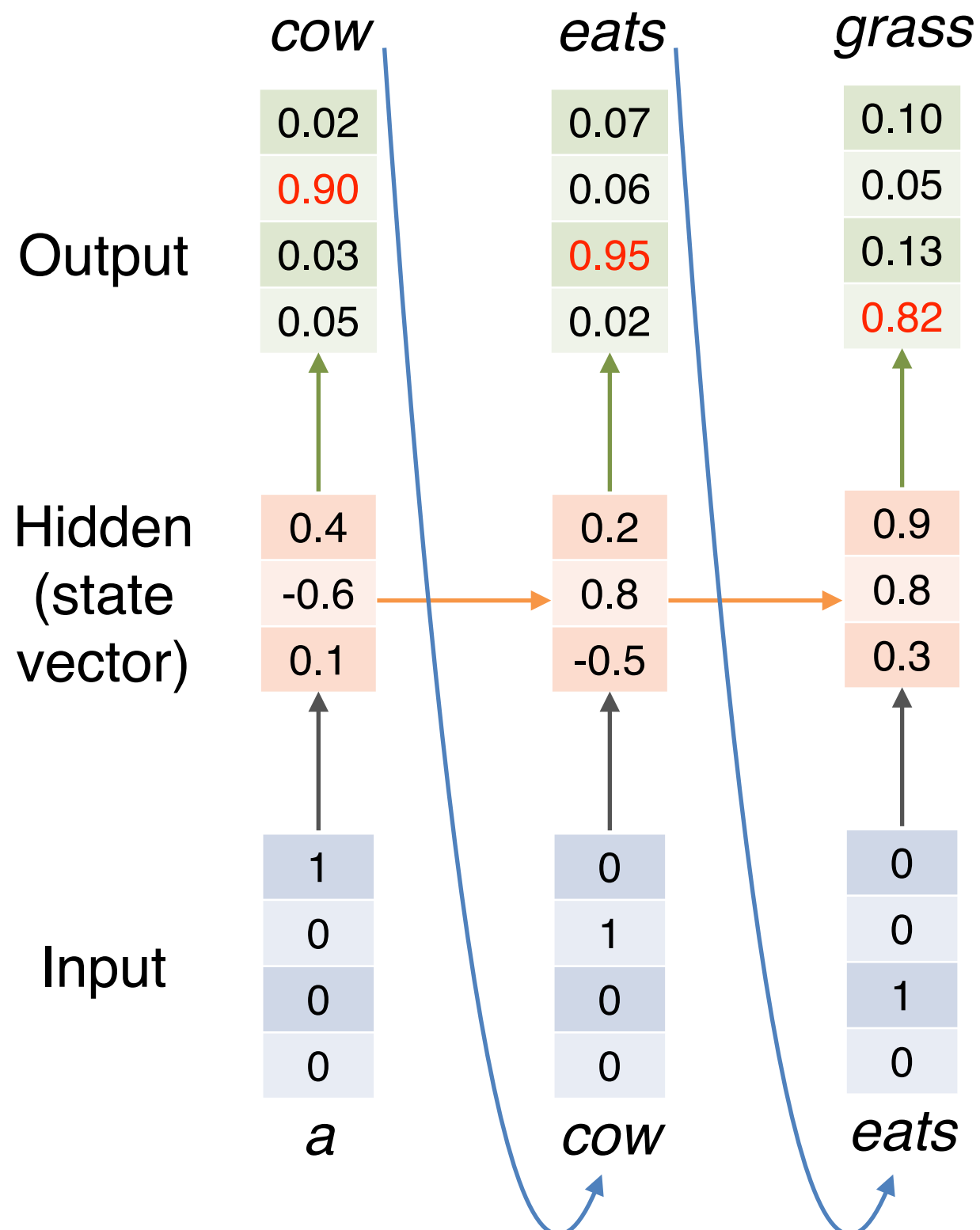
$$L_1 = -\log(0.30)$$

$$L_2 = -\log(0.50)$$

$$L_3 = -\log(0.20)$$

$$L_{\text{total}} = L_1 + L_2 + L_3$$

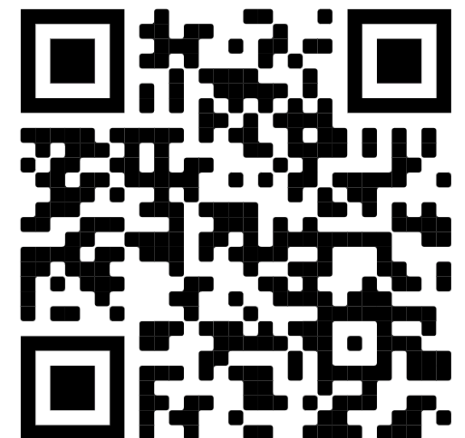
RNN Language Model: Generation



What Are Some Potential Problems with This Generation Approach?

- Mismatch between training and decoding
- Error propagation: unable to recover from errors in intermediate steps
- Tends to generate “bland” or “generic” language

PollEv.com/jeyhanlau569



Long Short-term Memory Networks

Language Model... Solved?

- RNN has the capability to model infinite context
- But can it actually capture long-range dependencies in practice?
- No... due to “**vanishing gradients**”
- Gradients in later steps diminish quickly during backpropagation
- Earlier inputs do not get much update

Long Short-term Memory (LSTM)

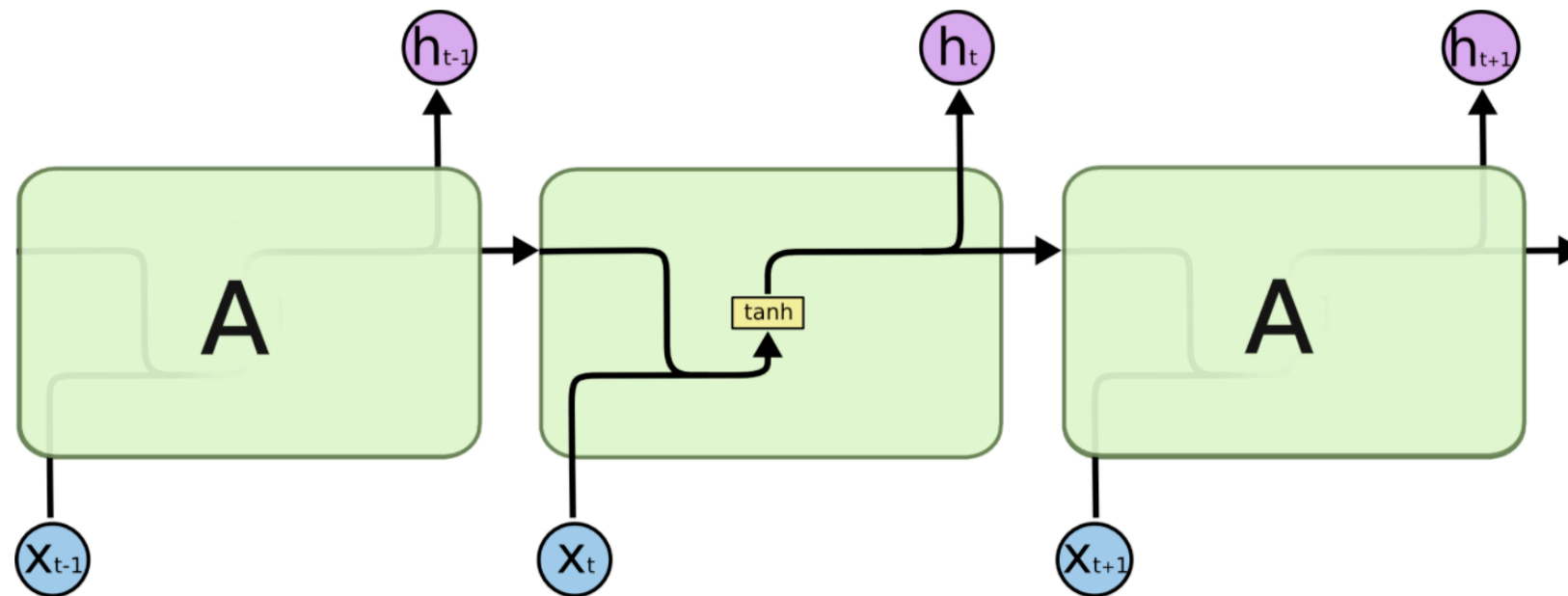
- LSTM is introduced to solve vanishing gradients
- Core idea: have “memory cells” that preserve gradients across time
- Access to the memory cells is controlled by “gates”
- For each input, a gate decides:
 - how much the new input should be written to the memory cell
 - and how much content of the current memory cell should be forgotten

Gating Vector

- A gate $\mathbf{g}$ is a vector
 - each element has values between 0 to 1
 - Use sigmoid function to produce $\mathbf{g}$
- $\mathbf{g}$ is multiplied component-wise with vector $\mathbf{v}$, to determine how much information to keep for $\mathbf{v}$

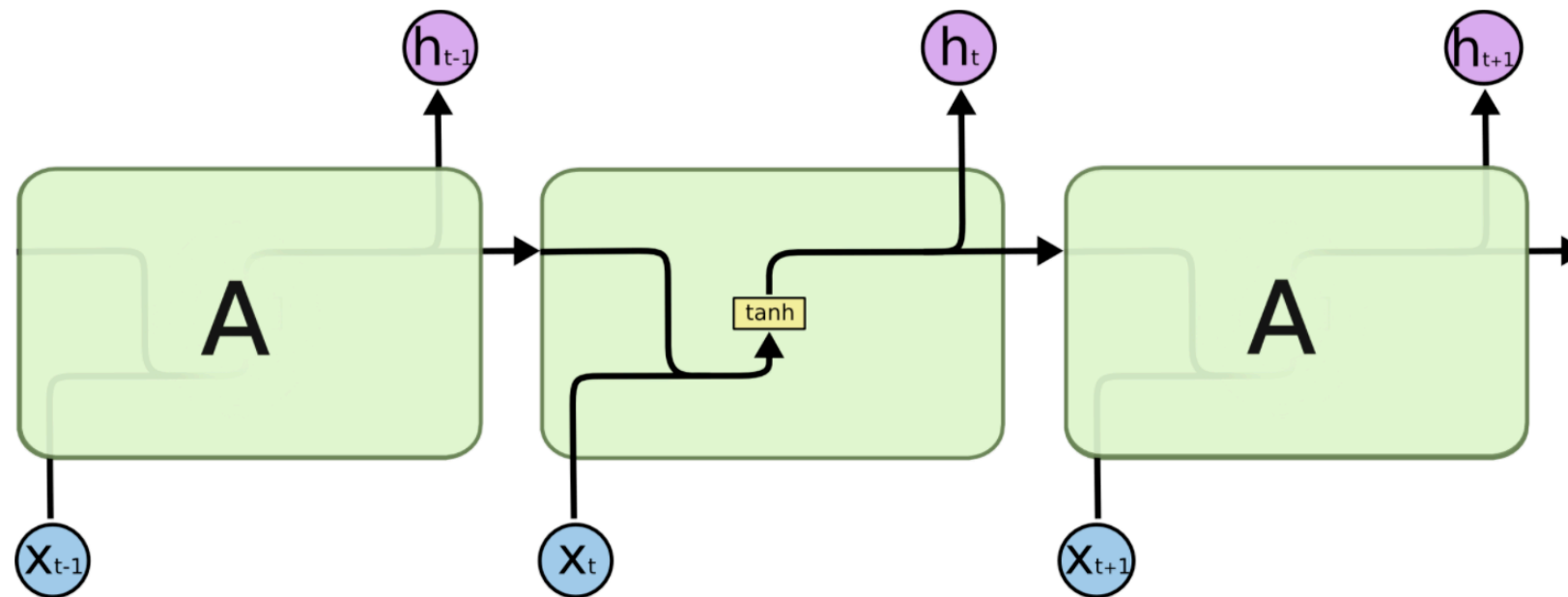
0.9		2.5		2.3
0.1	*	5.3	=	0.5
0.0		1.2		0.0
$\mathbf{g}$		$\mathbf{v}$		

Simple RNN vs. LSTM

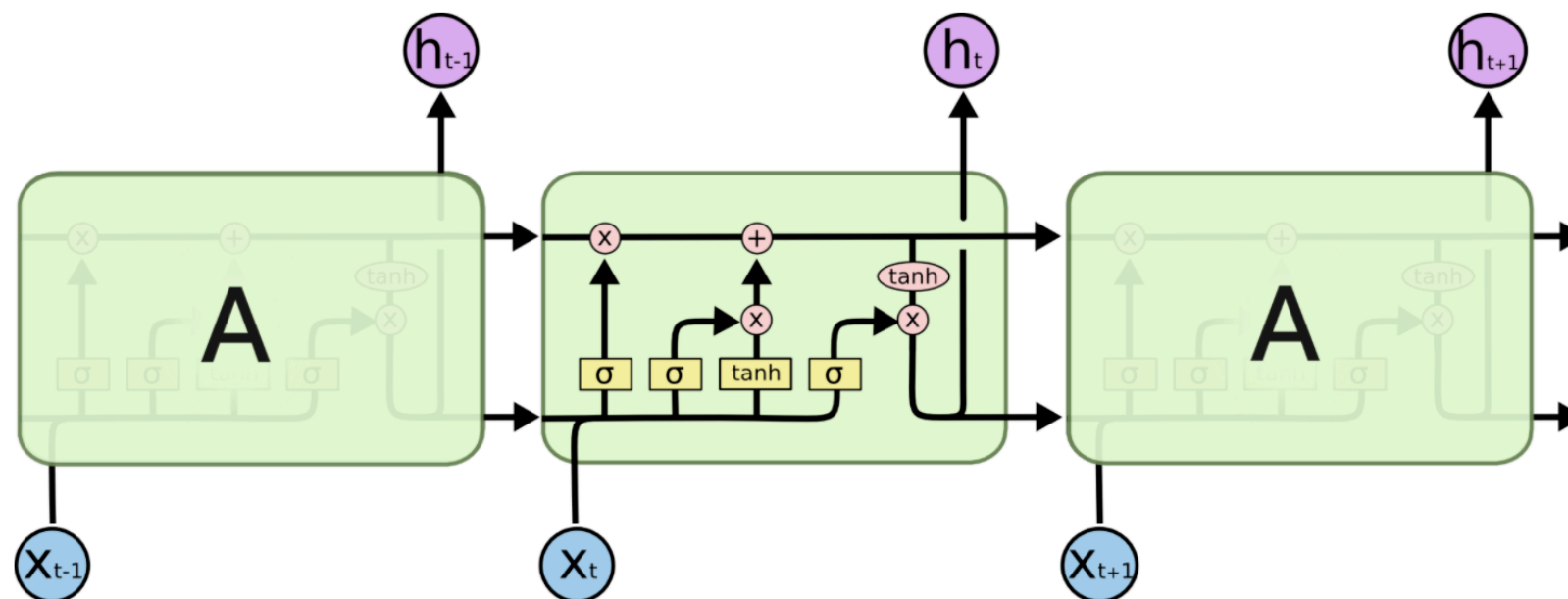


The repeating module in a standard RNN contains a single layer.

Simple RNN vs. LSTM

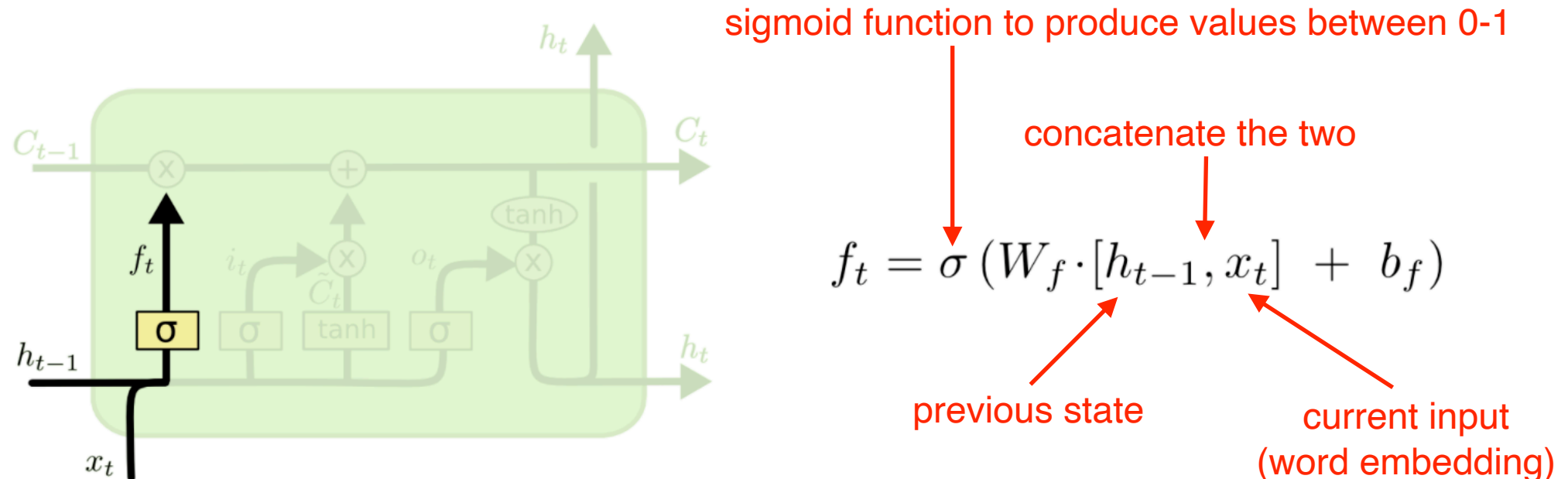


The repeating module in a standard RNN contains a single layer.



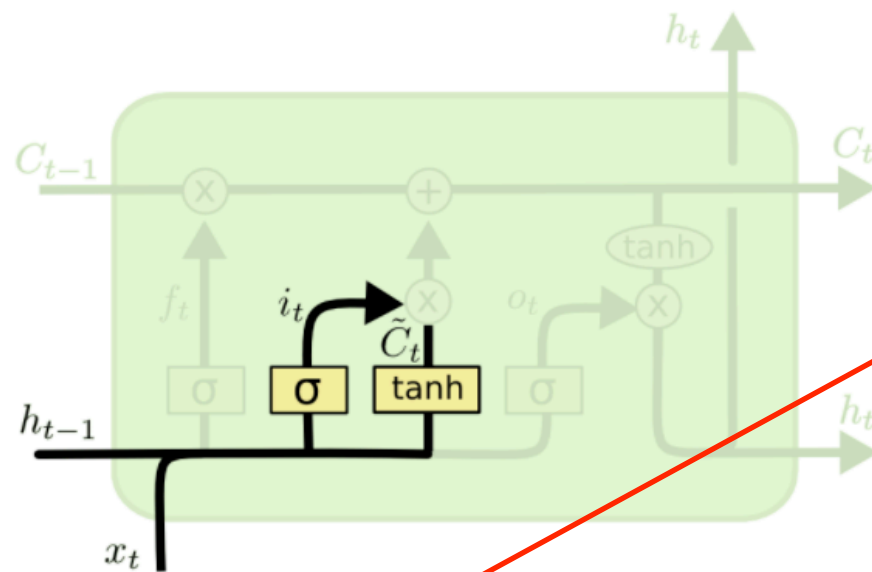
The repeating module in an LSTM contains four interacting layers.

LSTM: Forget Gate



- Controls how much information to “forget” in the memory cell (C_{t-1})
- The cats that the **boy** likes*
- Memory cell was storing noun information (*cats*)
- The cell should now forget *cats* and store *boy* to correctly predict the singular verb *likes*

LSTM: Input Gate

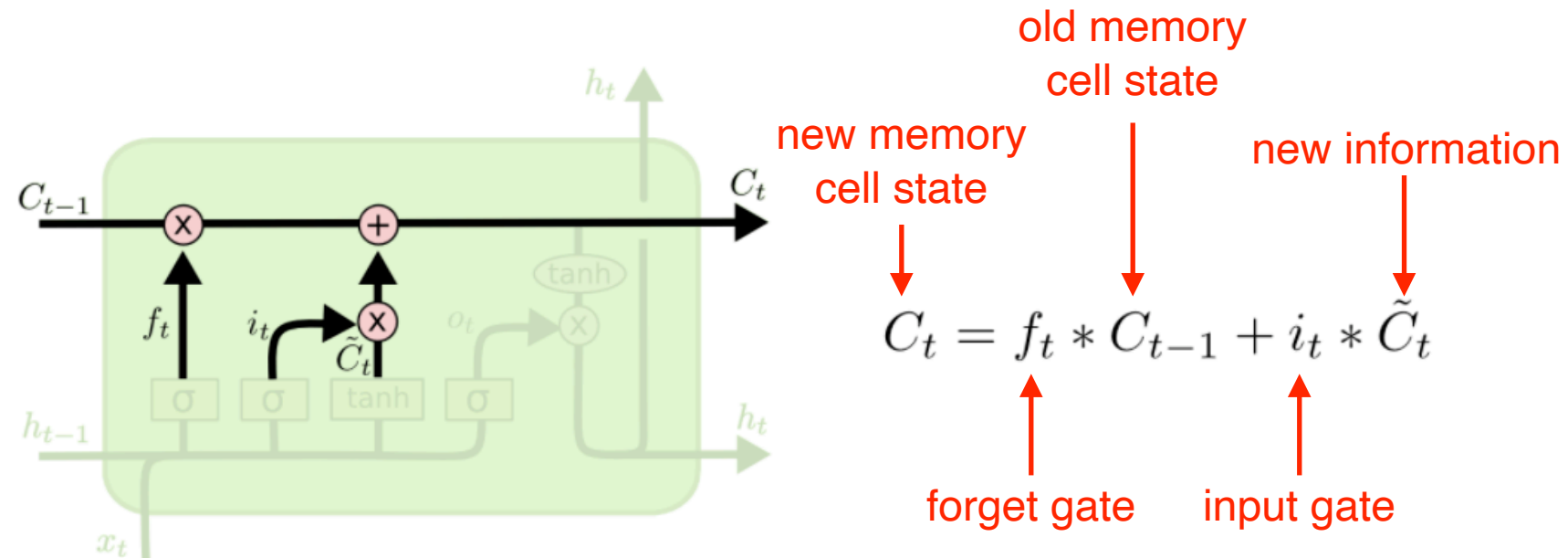


$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

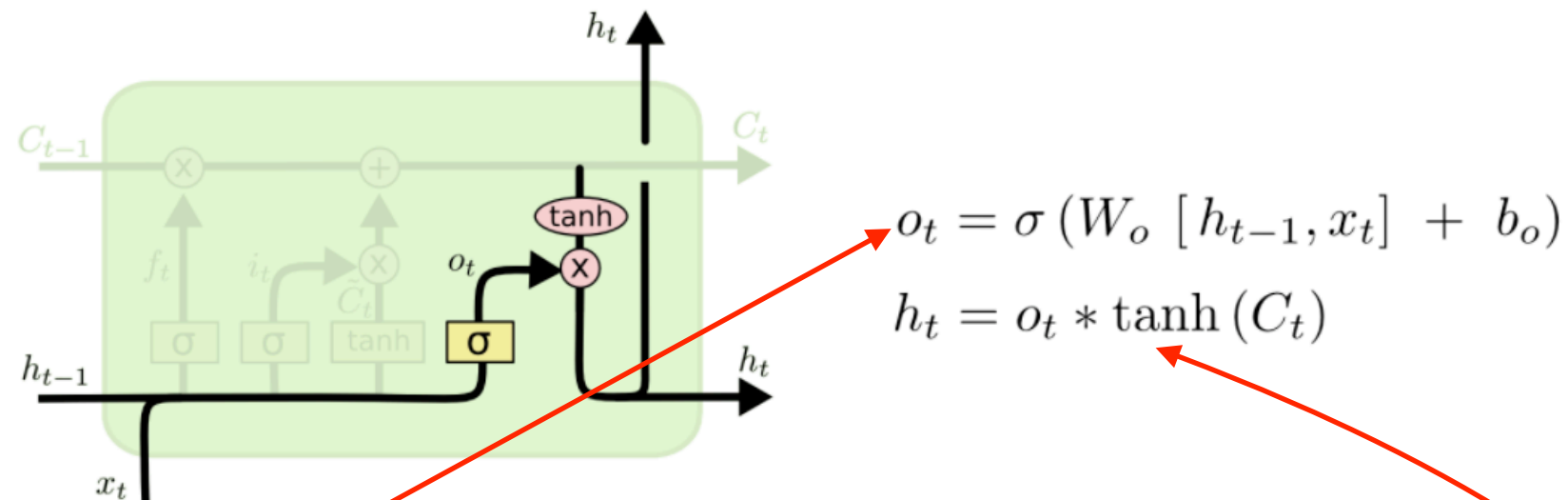
- Input gate controls how much new information to put to memory cell
- $\tilde{C}$ = new distilled information to be added
 - e.g. information about *boy*

LSTM: Update Memory Cell



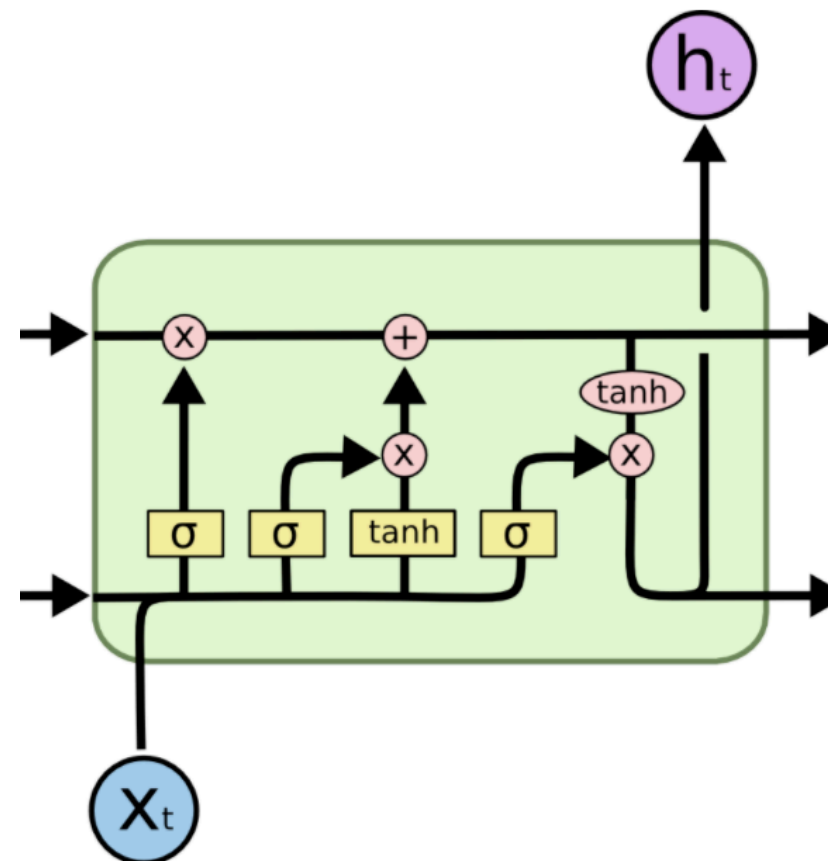
- Use the forget and input gates to update memory cell

LSTM: Output Gate



- Output gate controls how much to distill the content of the memory cell to create the next state (h_t)

LSTM: Summary



$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

$$h_t = o_t * \tanh(C_t)$$

Applications

Shakespeare Generator

PANDARUS:

Alas, I think he shall be come approached and the day
When little strain would be attain'd into being never fed,
And who is but a chain and subjects of his death,
I should not sleep.

Second Senator:

They are away this miseries, produced upon my soul,
Breaking and strongly should be buried, when I perish
The earth and thoughts of many states.

DUKE VINCENTIO:

Well, your wit is in the care of side and that.

Second Lord:

They would be ruled after this chamber, and
my fair nudes begun out of the fact, to be conveyed,
Whose noble souls I'll have the heart of the wars.

Clown:

Come, sir, I will make did behold your worship.

VIOLA:

I'll drink it.

- Training data = all works of Shakespeare
- Model: **character** RNN, hidden dimension = 512

Wikipedia Generator

Naturalism and decision for the majority of Arab countries' capitalide was grounded by the Irish language by [[John Clair]], [[An Imperial Japanese Revolt]], associated with Guangzham's sovereignty. His generals were the powerful ruler of the Portugal in the [[Protestant Immineners]], which could be said to be directly in Cantonese Communication, which followed a ceremony and set inspired prison, training. The emperor travelled back to [[Antioch, Perth, October 25|21]] to note, the Kingdom of Costa Rica, unsuccessful fashioned the [[Thrales]], [[Cynth's Dajoard]], known in western [[Scotland]], near Italy to the conquest of India with the conflict. Copyright was the succession of independence in the slop of Syrian influence that was a famous German movement based on a more popular servicious, non-doctrinal and sexual power post. Many governments recognize the military housing of the [[Civil Liberalization and Infantry Resolution 265 National Party in Hungary]], that is sympathetic to be to the [[Punjab Resolution]] (PJS)[<http://www.humah.yahoo.com/guardian.cfm/7754800786d17551963s89.htm> Official economics Adjoint for the Nazism, Montgomery was swear to advance to the resources for those Socialism's rule, was starting to signing a major tripad of aid exile.]]

- Training data = 100MB of Wikipedia raw data

Code Generator

```
/*
 * Increment the size file of the new incorrect UI_FILTER group information
 * of the size generatively.
 */
static int indicate_policy(void)
{
    int error;
    if (fd == MARN_EPT) {
        /*
         * The kernel blank will coeld it to userspace.
         */
        if (ss->segment < mem_total)
            unblock_graph_and_set_blocked();
        else
            ret = 1;
        goto bail;
    }
    segaddr = in_SB(in.addr);
    selector = seg / 16;
    setup_works = true;
    for (i = 0; i < blocks; i++) {
        seq = buf[i++];
        bpf = bd->bd.next + i * search;
        if (fd) {
            current = blocked;
        }
    }
    rw->name = "Getjbbregs";
    bprm_self_clearl(&iv->version);
    regs->new = blocks[(BPF_STATS << info->historidac)] | PFMR_CLOBATHINC_SECONDS << 12;
    return segtable;
}
```

Deep-Speare

- Generates Shakespearean sonnets

```
python sonnet_gen.py -m trained_model/ -d 1
```

```
Temperature = 0.6 - 0.8
```

```
01 [0.43] with joyous gambols gay and still array  
02 [0.44] no longer when he twas, while in his day  
03 [0.00] at first to pass in all delightful ways  
04 [0.40] around him, charming and of all his days
```

```
python sonnet_gen.py -m trained_model/ -d 2
```

```
Temperature = 0.6 - 0.8
```

```
01 [0.44] shall i behold him in his cloudy state  
02 [0.00] for just but tempteth me to stop and pray  
03 [0.00] a cry: if it will drag me, find no way  
04 [0.40] from pardon to him, who will stand and wait
```

Computers produce poetry by the metre

Mark Bridge, Technology Correspondent

Tuesday July 31 2018, 12:01am BST, The Times



Shakespeare's 12th sonnet, illustrated by John Gilbert
CULTURE CLUB/GETTY IMAGES

Sonnets written by a computer are superior to Shakespeare's in some respects — and the general public is unable to tell the difference between the two, researchers say.

Computer scientists have trained a “neural network” using 2,600 sonnets taken from a free online database and put the system to work



Emerging Tech

Move over, Shakespeare: This sonnet-writing A.I. is the poet we need

By Luke Dormehl
July 30, 2018



Can YOU spot the real Shakespearean sonnet? The AI learning how write its own poetry

- Researchers at IBM used 2600 sonnets to train their AI
- Researchers asked workers from a crowdsourcing website to rate them
- Found they were unable to distinguish them from the real deal
- However, experts were more critical and said they still lacked real emotion

By [MARK PRIGG FOR DAILYMAL.COM](#)

PUBLISHED: 05:25 AEDT, 28 July 2018 | **UPDATED:** 05:56 AEDT, 28 July 2018



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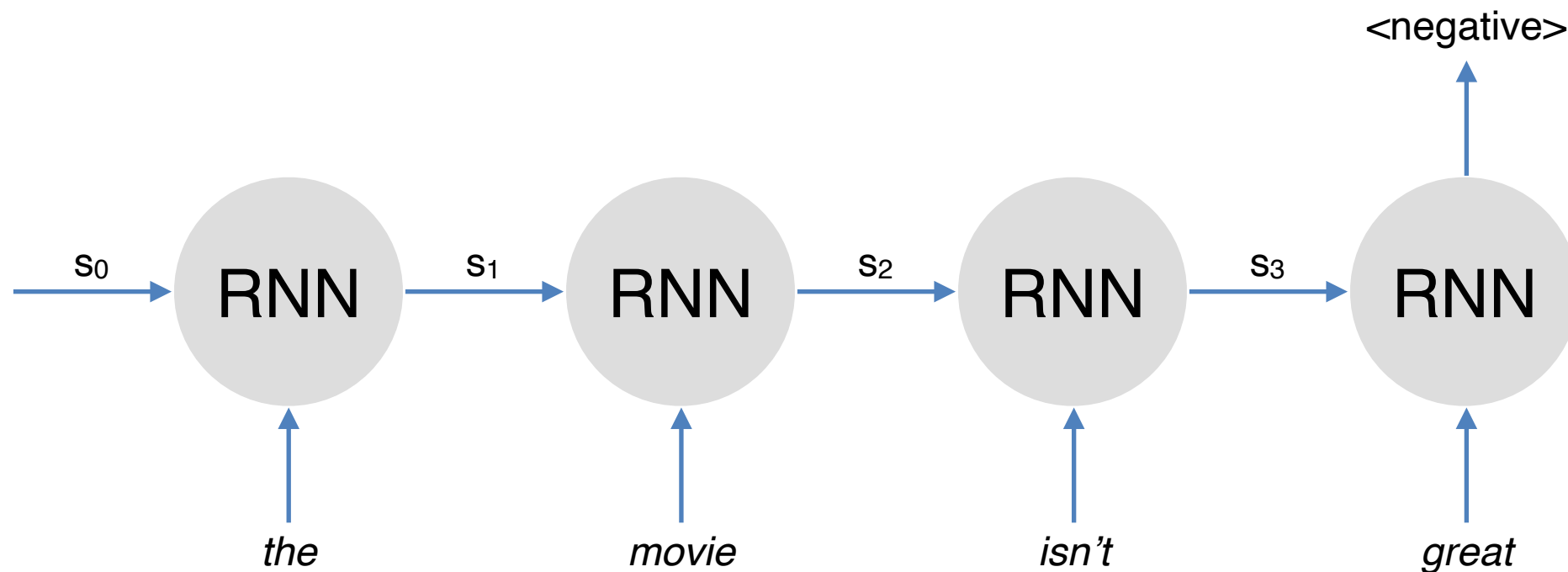
Researchers have revealed an AI that they hope will one day write a perfect Shakespearean sonnet.

Researchers at IBM used created a sonnet writing AI.

They trained it using 2600 sonnets taken from a free online database of out-of-copyright books called Project Gutenberg.

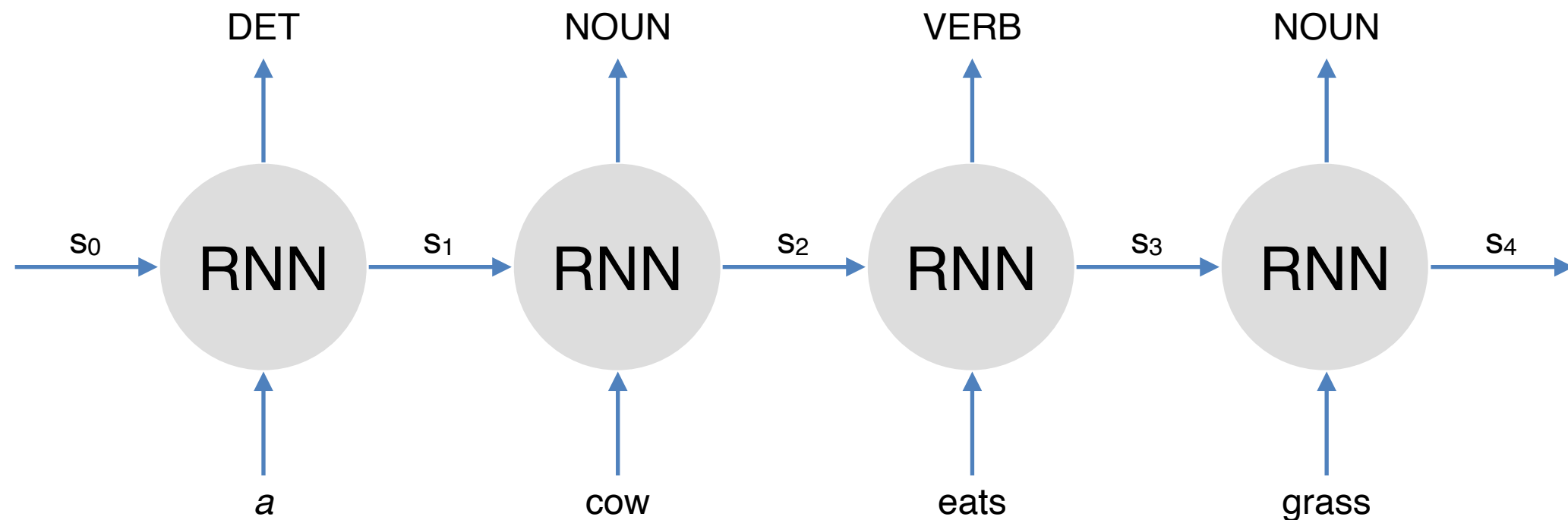
Text Classification

- RNNs can be used in a variety of NLP tasks
- Particularly suited for tasks where order of words matter, e.g. sentiment classification

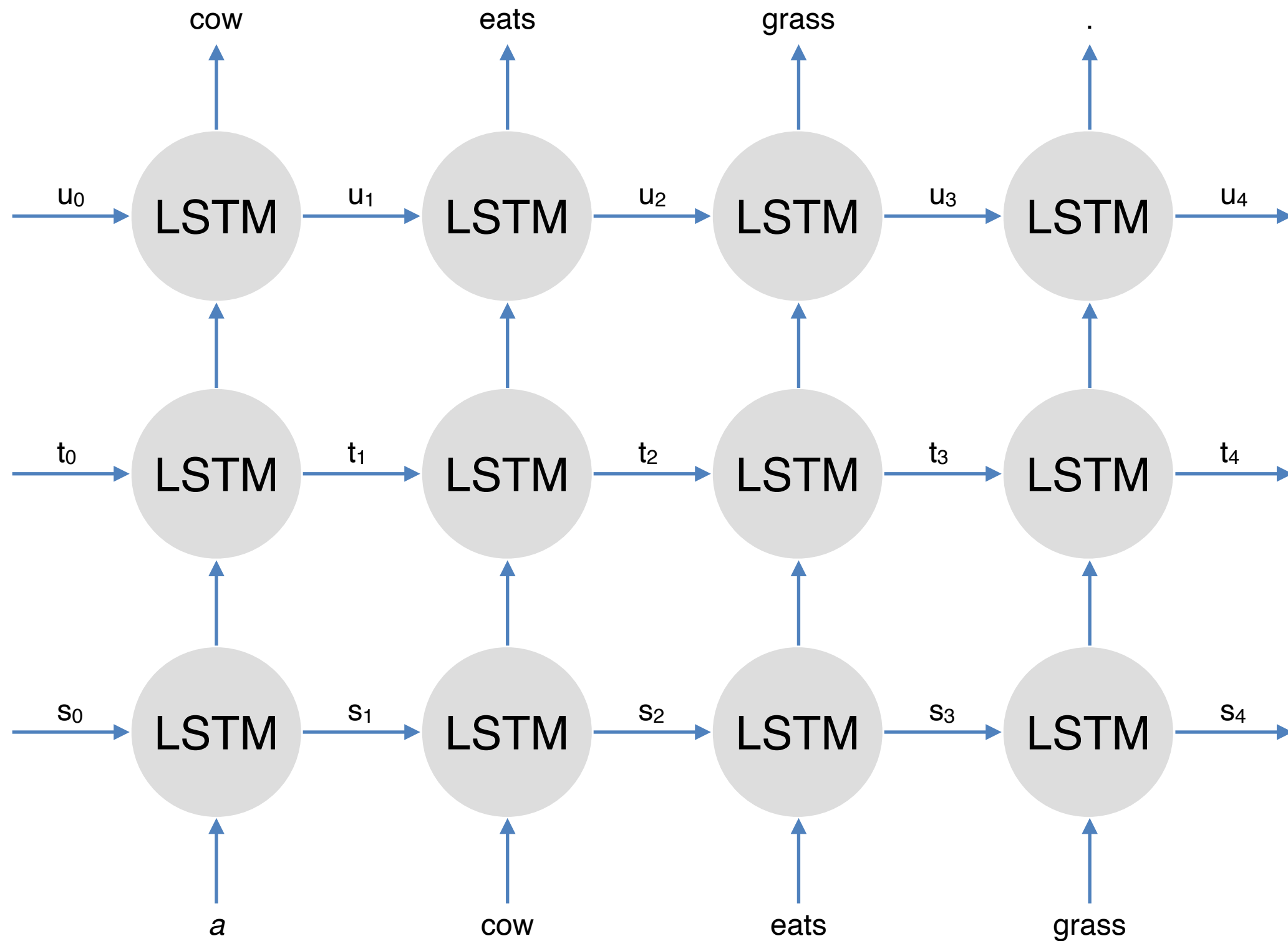


Sequence Labeling

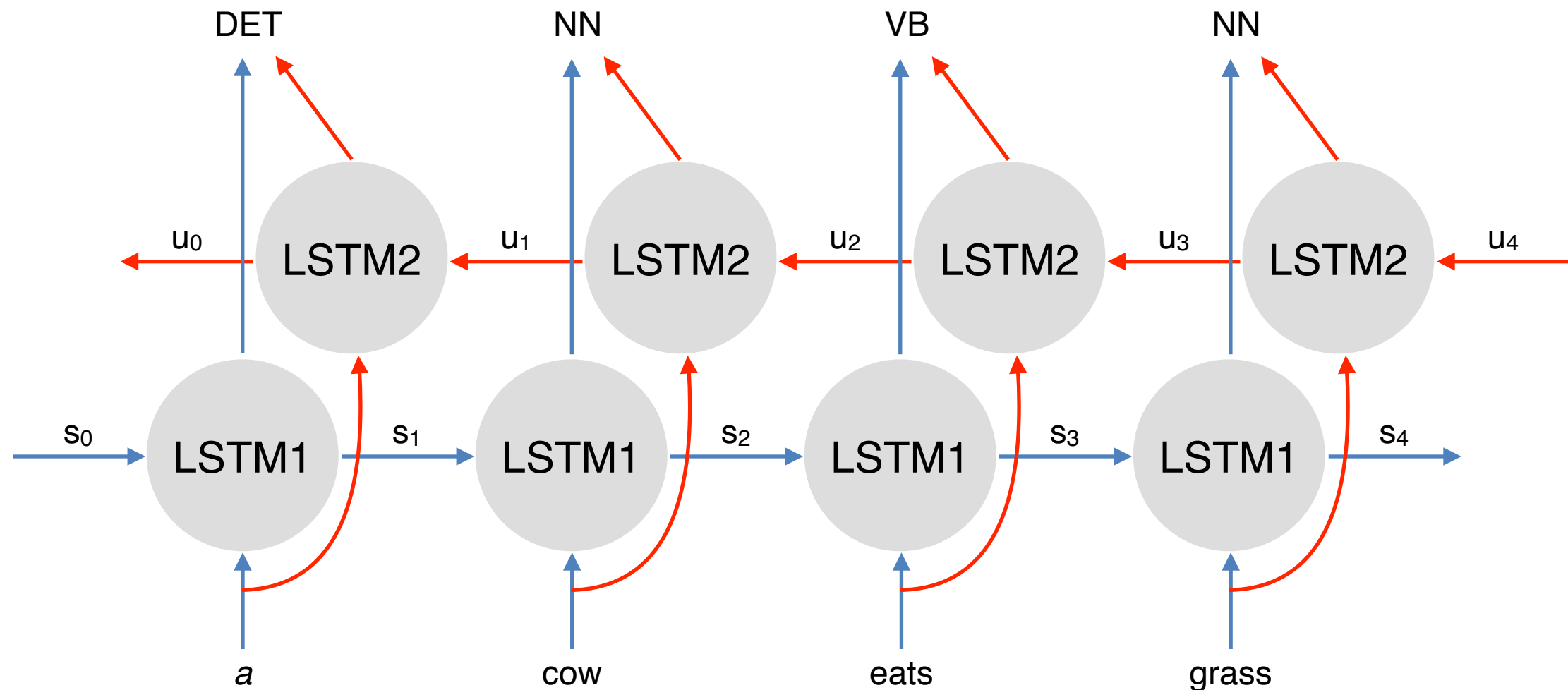
- Also good for sequence labelling problems, e.g. POS tagging



Multi-layer LSTM



Bidirectional LSTM



$$y_i = \text{softmax}(W_s[s_i, u_i])$$

use information from both
forward and backward LSTM

Final Words

- Pros
 - Has the ability to capture long range contexts
 - Just like feedforward networks: flexible
- Cons
 - Slower than FF networks due to sequential processing
 - In practice doesn't capture long range dependency very well (evident when generating very long text)
 - In practice also doesn't stack well (multi-layer LSTM)
 - Less popular nowadays due to the emergence of more advanced architectures (Transformer!)

Readings

- JM3 Chapter 13.1-13.5